POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Business Case 2
Group 12

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1. Introduction

Financial institutions increasingly rely on data-driven approaches to deliver personalized services and improve customer engagement. In this context, understanding clients' underlying financial needs is crucial, as these needs represent the primary drivers of product demand and investment behavior.
In this project, we model the estimation of financial needs as a supervised machine learning problem using implicit labeling.
The objective is to build predictive models capable of identifying clients' needs based on their characteristics, enabling more effective targeting and supporting the development of Next Best Action recommendation systems.
Particular attention is devoted to handling class imbalance and optimizing recall, as missing a relevant client need can have significant business implications.

2. Business Framework

Financial needs are not directly observable and must be inferred from available data. From what we experience, we can say that needs depend on the client's characteristics and context:

$$Y = f(X)$$

where X represents client features and Y the presence of a specific need.
Since explicit labels are not available, needs are approximated using **implicit labels**: a client is assumed to have a need if they own a product satisfying it. This approach aligns with real-world financial advisory processes, where observed behavior proxies latent needs.

3. Dataset and Preprocessing

The dataset contains client-level information, including demographic characteristics, financial conditions and behavioral indicators. These variables are used to approximate the economic situation and preferences of each client.
The preprocessing phase is designed to ensure data quality, improve statistical properties of the variables, and enhance model performance. The main steps are:

- **Data cleaning and missing values**: inconsistent entries are corrected and missing values are handled through appropriate imputation strategies, ensuring a coherent and usable dataset.

- **Transformation of skewed variables**: highly skewed financial variables, such as income and wealth, are transformed using logarithmic scaling to reduce asymmetry, stabilize variance, and improve model interpretability, since from a financial perspective, proportional differences are more informative than absolute ones.
- **Feature engineering**: additional variables are constructed to capture non-linear effects and economic relationships between variables, particularly interactions between financial resources, risk propensity, and household structure.
- **Target construction**: binary target variables are defined for each financial need using an implicit labeling approach, where product ownership is used as a proxy for the presence of a need.

This preprocessing pipeline ensures that the input features are informative, well-behaved from a statistical perspective, and suitable for downstream machine learning models.

4. Feature Engineering

To enhance the predictive power of the models, several derived features were introduced to better capture the economic conditions and behavioral characteristics driving financial needs.

- **Income-Wealth Ratio**: this feature captures the balance between income and accumulated wealth. It provides insight into whether a client is in an accumulation phase (high income relative to wealth) or in a decumulation/stability phase, which is directly linked to different financial needs.
- **Risk interaction terms**: these variables model the interaction between financial capacity, financial literacy and risk propensity. They aim to capture heterogeneous behavior, as clients with similar wealth levels may exhibit very different investment needs depending on their risk tolerance and financial knowledge.
- **Wealth per family member**: this variable acts as a proxy for financial pressure and resource allocation within the household. Lower values may indicate tighter constraints and a higher likelihood of protection or liquidity-related needs.
- **Retirement proximity**: this feature captures the life-cycle dimension of financial decision-making. As clients approach retirement, their needs typically shift from accumulation to income generation, capital preservation, and decumulation strategies.

However, the inclusion of these features led to only marginal improvements in model performance, particularly when targeting IncomeInvestment. This suggests that, while economically meaningful, the available variables may not fully capture the complexity of clients' needs, highlighting the presence of significant unobserved factors and noise in the labeling process.

5. Modeling Approach

The estimation of financial needs is formulated as a supervised classification problem. Given the multi-label nature of the task (each client may exhibit multiple needs simultaneously), a **One-vs-All** strategy is adopted. For each model, the dataset is split into training and test sets, and predictions are expressed in terms of probabilities, representing the likelihood that a given client exhibits a specific need. These probabilities are later converted into binary outcomes through threshold optimization.

Several modeling techniques with increasing levels of complexity are implemented:

- **Logistic Regression**: used as a baseline model due to its simplicity and interpretability. It provides a linear benchmark and allows for an initial assessment of feature relevance.
- **Random Forest**: a method that captures non-linear relationships and interactions between variables while reducing variance through aggregation.
- **XGBoost**: a gradient boosting algorithm that sequentially improves model performance by focusing on misclassified observations.
- **Neural Networks**: employed to model potentially complex, non-linear patterns in the data through multiple layers of representation.

Hyperparameter tuning for the more complex models (Random Forest, XGBoost, and Neural Networks) is performed using an automated optimization framework based on **Optuna**.

The optimization process explores the hyperparameter space using a guided search strategy, with the objective of maximizing the F1-score on validation data. This approach allows for a systematic and efficient calibration of model parameters, improving performance while avoiding manual tuning.

However, despite the increasing model complexity, performance improvements remain limited.

5.1. Ensemble and Soft Encoding

To further improve predictive performance and reduce model-specific bias, an ensemble approach is implemented by combining the probabilistic outputs of different models.
Instead of relying on a single classifier, the predicted probabilities for each need are aggregated across models through a soft combination scheme. This approach leverages the complementary strengths of different algorithms, smoothing individual model errors and improving the robustness of the final predictions.
This results in a **soft encoding** of client needs, where each need is represented by a continuous score reflecting the consensus across models, rather than a binary decision from a single model.
This representation is particularly suitable for the subsequent recommendation stage, as it provides a more stable and informative ranking of needs and mitigates overfitting to specific model assumptions.

6. Evaluation Metrics

Model performance is evaluated using precision, recall, F1-score, and accuracy. The primary metric for model selection is the **F1-score**, as it provides a balanced trade-off between precision and recall.
This choice avoids degenerate solutions where the model predicts predominantly one class (all positives or all negatives), which could artificially inflate recall or precision while providing little practical value.

- **Recall**: measures the ability to correctly identify clients with a given need. It is the key metric, as failing to detect a true need directly translates into missed business opportunities.
- **Precision**: evaluates the proportion of correctly identified needs among all predicted positives, reflecting the reliability of the model's recommendations.
- **F1-score**: provides a weighted average between precision and recall, ensuring that improvements in recall do not come at the cost of excessively low precision.
- **Accuracy**: measures the overall proportion of correct predictions. However, due to class imbalance, it may provide a misleading indication of performance, as high accuracy can be achieved by simply predicting the majority class.

Particular attention is devoted to maintaining a reasonable trade-off between recall and precision.

7. Threshold Optimization

Predicted probabilities are converted into binary decisions through threshold tuning. Instead of using the default cutoff of 0.5, thresholds are optimized to better align model outputs with business objectives.
In particular, thresholds are selected to:

- maximize F1-score
- maintain an acceptable level of precision
- preserve a balanced F1-score

This procedure allows explicit control over the trade-off between false positives and false negatives, which is essential in the presence of class imbalance and asymmetric business costs.

8. Results

Overall, the models exhibit moderate predictive performance, with recall values typically around 0.5 for some targets. Increasing model complexity, feature engineering and threshold optimization do not lead to substantial improvements, suggesting that the main limitation is not the choice of algorithm but the quality of the input data and target construction.
A relevant difference emerges across target variables. In particular, **Accumulation Investment** tends to be predicted more effectively than **Income Investment**. This can be explained by the stronger relationship between accumulation needs and observable features such as income level, wealth, age, and life-cycle position.
In contrast, **Income Investment** is more difficult to predict, as it depends on more complex and less observable factors, such as individual preferences for income stability, risk aversion, and specific financial goals. These aspects are only partially captured by the available features, leading to weaker model performance.
More generally, the main limitation lies in the **quality of labels**. Since needs are inferred indirectly from product ownership, the resulting targets are noisy and may not accurately reflect the truth. This issue is compounded by several factors:

- product ownership does not necessarily imply the presence of a need
- some needs may exist without being translated into observable products

- static features fail to capture dynamic and behavioral aspects of decision-making
- class imbalance further limits the ability of models to learn minority patterns

These results highlight that the primary bottleneck is the representation of needs, rather than the modeling technique, and that improvements are more likely to come from better data and labeling strategies than from increased model complexity.

9. Recommendation Framework

The estimated probabilities of each financial need are used to construct a recommendation layer within a Next Best Action framework.

For each client, the output of the classification models is a vector of predicted probabilities, one for each need. These probabilities are then used to rank needs in descending order, allowing the identification of the most relevant ones for each individual. Rather than relying on a single binary decision, this ranking-based approach preserves more information and enables a more flexible recommendation strategy.

The top-ranked needs are subsequently mapped to corresponding financial products, based on a predefined association between needs and product categories. This mapping allows the system to translate model predictions into recommendations. The recommendation process combines both threshold-based filtering and ranking: thresholds are used to discard low-confidence predictions, while the ranking ensures prioritization among the remaining needs. This hybrid approach avoids over-recommending irrelevant products while still capturing multiple potential needs for each client. The quality of recommendations is directly linked to the predictive performance of the underlying models. In particular, stronger results are observed for needs such as **Accumulation Investment**, where predictions are more reliable and aligned with observable financial characteristics.

Conversely, recommendations related to **Income Investment** are less accurate, reflecting the lower predictive performance of the corresponding models.

Overall, the recommendation system provides a coherent ranking of client needs, but its effectiveness is constrained by the same limitations affecting the predictive models, that can be improved by applying some modifications in both data representation and target construction.

First, the inclusion of richer data sources, such as transactional or time-series information, would allow the models to capture dynamic and behavioral patterns that are currently unobserved. These aspects play a key role in explaining financial needs, particularly for more complex targets.

Second, refining the labeling strategy is crucial. Incorporating expert knowledge, filtering biased observations, or combining multiple labeling approaches could lead to more reliable targets.

Finally, adopting learning strategies explicitly aligned with the business objective, such as cost-sensitive methods or recall-oriented optimization, could improve the identification of clients with relevant needs without excessively compromising precision.